

SATELLITE HEALTH MONITORING SYSTEM USING ESP32

ABSTRACT

Satellite Health Monitoring System is an IoT-based embedded system designed to continuously monitor the health parameters of a satellite. The system uses an ESP32 microcontroller to collect data from sensors such as DHT11 and MPU6050. Temperature, humidity, acceleration, gyroscope, pitch, and roll values are monitored in real time. The collected data is displayed on an OLED display and transmitted to the ThingSpeak cloud platform for remote monitoring. A buzzer and LED provide immediate alerts whenever the temperature exceeds the predefined threshold. This project demonstrates the concept of satellite telemetry, fault detection, and health monitoring using low-cost hardware components.

INTRODUCTION

Satellites operate in harsh environments and require continuous monitoring of critical parameters to ensure reliable operation. Health monitoring systems are used to observe the condition of onboard electronics and detect anomalies before failures occur. This project simulates a satellite health monitoring system using ESP32 and various sensors. The system acquires environmental and motion-related data and displays it locally as well as on a cloud dashboard.

OBJECTIVES

1. To monitor temperature and humidity using DHT11.
2. To measure acceleration and angular motion using MPU6050.
3. To calculate pitch and roll orientation.
4. To display sensor data on an OLED screen.
5. To upload telemetry data to ThingSpeak cloud.
6. To generate visual and audible alerts during abnormal conditions.
7. To simulate a real-time satellite health monitoring system.

COMPONENTS USED

S.No	Component	Quantity
1	ESP32 Development Board	1
2	MPU6050 Accelerometer & Gyroscope	1
3	DHT11 Temperature & Humidity Sensor	1
4	0.96" OLED Display (SSD1306)	1
5	Active Buzzer	1

S.No	Component	Quantity
6	LED Indicator	1
7	220Ω Resistor	1
8	Jumper Wires	As Required
9	USB Cable	1
10	Breadboard	1

HARDWARE DESCRIPTION

ESP32

ESP32 is a powerful microcontroller with built-in Wi-Fi and Bluetooth capabilities. It acts as the central processing unit of the project.

DHT11 Sensor

The DHT11 sensor measures temperature and humidity. It is used to monitor the internal environmental conditions of the simulated satellite.

MPU6050 Sensor

The MPU6050 is a six-axis sensor containing a three-axis accelerometer and a three-axis gyroscope. It measures movement, vibration, and orientation.

OLED Display

The OLED display presents sensor readings and system status in real time.

Buzzer and LED

The buzzer and LED provide warning indications when the temperature exceeds the safe operating threshold.

SOFTWARE USED

- Arduino IDE
- ESP32 Board Package
- ThingSpeak Cloud Platform
- C/C++ Programming Language

WORKING PRINCIPLE

1. ESP32 initializes all sensors and connects to Wi-Fi.

2. DHT11 measures temperature and humidity.
3. MPU6050 measures acceleration and gyroscope values.
4. Pitch and roll angles are calculated using acceleration data.
5. Sensor values are displayed on the OLED screen.
6. Data is transmitted to ThingSpeak cloud at regular intervals.
7. If temperature exceeds 35°C, the buzzer and LED are activated.
8. The system continuously monitors and updates all parameters.

OLED DISPLAY SCREENS

Screen 1

- Temperature
- Humidity
- Pitch
- Roll

Screen 2

- Satellite Health Status

Screen 3

- Acceleration Values
- AX, AY, AZ

Screen 4

- Gyroscope Values
- GX, GY, GZ

THINGSPEAK FIELDS

Field	Parameter
Field 1	Temperature
Field 2	Humidity
Field 3	Acceleration X
Field 4	Acceleration Y
Field 5	Acceleration Z
Field 6	Gyroscope X
Field 7	Gyroscope Y
Field 8	Gyroscope Z

RESULTS

The system successfully monitors temperature, humidity, acceleration, and gyroscope data. Sensor readings are displayed on the OLED screen and uploaded to the ThingSpeak cloud platform. The alert mechanism successfully activates when the temperature exceeds the threshold value.

APPLICATIONS

- Satellite Telemetry Systems
- Spacecraft Health Monitoring
- Aerospace Systems
- Remote Environmental Monitoring
- Industrial Equipment Monitoring
- IoT-Based Condition Monitoring

ADVANTAGES

- Real-time monitoring
- Wireless cloud connectivity
- Low-cost implementation
- Easy fault detection
- Portable and scalable design

FUTURE ENHANCEMENTS

1. Addition of voltage and current sensors.
2. Integration of vibration monitoring sensors.
3. GPS-based location tracking.
4. Advanced anomaly detection using AI.
5. Real-time mobile application monitoring.
6. Data logging using SD card storage.

CONCLUSION

The Satellite Health Monitoring System using ESP32 successfully demonstrates the concept of real-time telemetry and condition monitoring. The project integrates multiple sensors, cloud connectivity, and alert mechanisms to simulate a basic satellite monitoring system. It provides a strong foundation for advanced aerospace and embedded system applications.

REFERENCES

1. ESP32 Datasheet
2. MPU6050 Datasheet
3. DHT11 Datasheet
4. ThingSpeak Documentation
5. Arduino IDE Documentation